

# Lecture 2: The Working of Competitive Markets

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# Lecture overview

- Business in a Competitive Market
- The Working of Competitive Markets – Demand
- The Working of Competitive Markets – Supply

# What is the Best Price?

- The "best" price is subjective and depends entirely on a firm's specific objectives.
- Primary Commercial Objectives:
  1. Profit Maximization: Achieving the highest possible surplus over costs.
  2. Market Share Maximization: Capturing the largest percentage of total consumers in the industry.
  3. Revenue Maximization: Generating the highest total "top-line" cash flow.

The Trade-off: Objectives often conflict. For example, lowering prices to boost market share may sacrifice profit margins.

# The Pricing Dilemma: High Volume vs. High Margin

- Firms must navigate the inverse relationship between price and sales:
  - Low Price: Generates high sales volume but lower revenue per unit.
  - High Price: Generates high margins but risks low sales volume.
- **Example:** a £5 price (10 sales) and a £10 price (5 sales) both result in £50 revenue. Is best price £8 or 8.25?
- For massive retailers like Amazon, even a tiny price adjustment (e.g., a few pence) can lead to massive shifts in total revenue due to high sales volumes.
- Modern pricing is no longer guesswork. Every search, click, and purchase provides data that firms model to determine strategic, real-time price changes.

# The Firm and the Market Environment



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- The market environment is often outside a firm's direct control and subject to frequent change.
- A firm's strategy regarding prices and output depends heavily on the state of the market.
  - **Buoyant Market:** Firms may increase output and raise prices in anticipation of higher consumer willingness to pay.
  - **Declining Market:** Firms may be forced to reduce output, cut prices, or diversify into new products.

# Price Takers and Market Power

- **Firms as Price Takers:** In highly competitive markets, firms must accept the market price as given; they have no power to change it.
  - Example: Farmers selling wheat must accept the market rate; if they raise prices individually, they will lose all sales to competitors.
- **Consumers as Price Takers:** In competitive retail environments, consumers also have no control over prices and must accept them as given
  - Example: you cannot haggle at a supermarket checkout.
- **Perfect Competition:** A theoretical situation where there are so many producers and consumers that everyone is a price taker.
- **Market Power:** In less competitive markets, some firms (like Ford) have the "market power" to exercise discretion over the prices they charge.

# The Price Mechanism and Equilibrium



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- Free Market Decisions: Consumers are free to make demand decisions, while firms are free to make supply decisions.
- Prices act as the communication tool between firms and consumers by responding to shortages and surpluses.
  - Shortages (Demand > Supply): Prices rise, encouraging firms to produce more for profit while discouraging consumer spending.
  - Surpluses (Supply > Demand): Prices fall, acting as a disincentive for producers to supply more while encouraging consumers to buy.
- Equilibrium Price: The point where demand exactly equals supply and the market is in balance.
- Price changes act as signals to move resources toward more profitable, high-demand goods and away from less profitable ones.

# Market Dynamics – Responding to Change

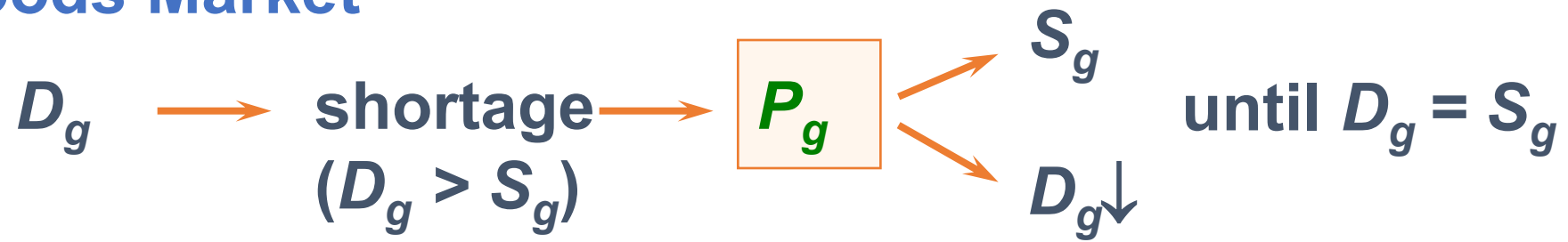


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- Goods Market Response to Increased Demand:
  - A rise in demand creates an immediate shortage.
  - This shortage causes the price of the good to rise.
  - The higher price eliminates the shortage by "choking off" some demand and incentivizing firms to increase production.

# The price mechanism: the effect of a rise in demand

## Goods Market



# Market Dynamics – Responding to Change

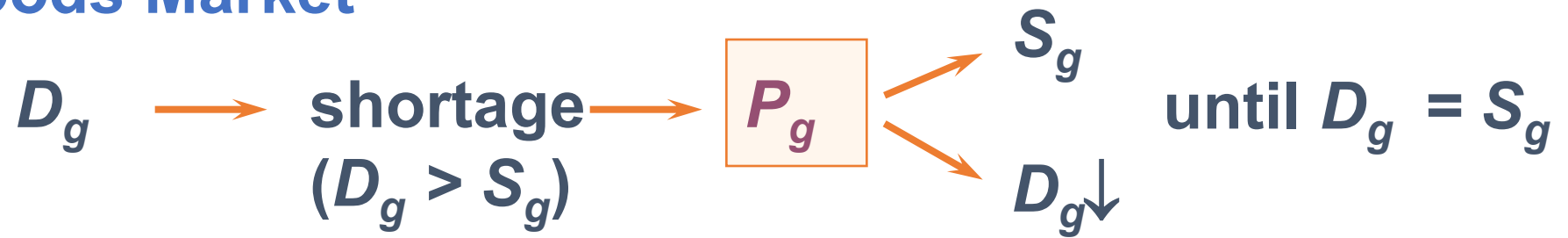


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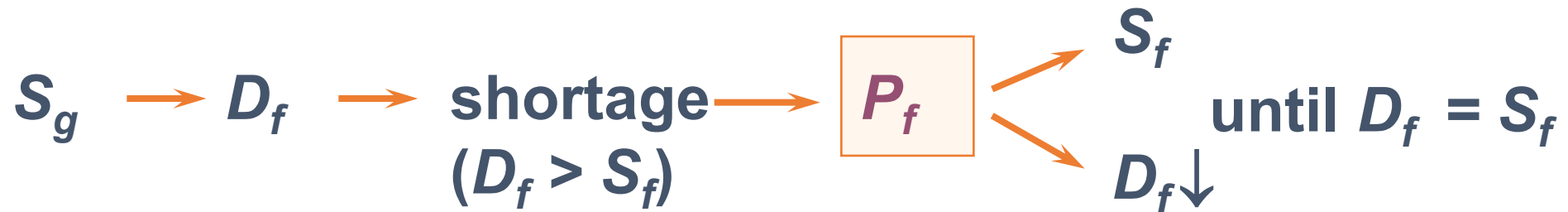
- Impact on Inputs (Factor Markets):
  - To increase supply, firms require more inputs (land, labour, capital).
  - This drives up the demand for inputs, subsequently raising the price of inputs.
  - Input suppliers respond to these higher prices by supplying more, eventually allowing firms to meet the higher consumer demand.

# The price mechanism: the effect of a rise in demand

## Goods Market



## Factor Market



# Market Dynamics – Responding to Change



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## How Factor Markets Affect Goods Markets:

- **Cost of Production:** Changes in input availability (e.g., discovering new raw materials) lower input prices.
- **Supply Shift:** Lower costs allow firms to increase the supply of finished goods.
- **Equilibrium Adjustment:** The resulting surplus of finished goods lowers their price, encouraging more consumer spending.

**Q** Assume that there is a fall in demand for a good. *Ceteris paribus*, this will result in:

- A. A *rise* in the price of the good and a *fall* in the price of factors used to make it.
- B. A *rise* in the price of the good and a *rise* in the price of factors used to make it.
- ☒ C. A *fall* in the price of the good and a *fall* in the price of factors used to make it.
- D. A *fall* in the price of the good and a *rise* in the price of factors used to make it.

# Interdependence Between Different Goods



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- Markets do not exist in isolation; the price of one good directly impacts the demand for another.
- If the price of a specific good rises, consumers look for alternatives.
  - This shift increases the demand for those alternatives.
  - The higher demand for alternatives drives their prices up, incentivizing producers to supply more of the substitute products.

# The Working of Competitive Markets - Demand

# The Law of Demand

- There is an inverse relationship between price and quantity demanded.
  - Example: If a coffee crop failure in Brazil causes shop prices to double, consumers will naturally cut back or switch to tea.
- The Two Drivers of the Law of Demand:
  - **The Income Effect:** As prices rise, consumers' "real income" (purchasing power) drops. They simply cannot afford to buy as much as they used to.
  - **The Substitution Effect:** As a good becomes more expensive relative to others, consumers switch to cheaper substitutes (e.g., switching from coffee to cocoa or fruit juice).
- Quantity Demanded is not just a "wish list." It refers only to the amount consumers are willing and able to purchase at a specific price.

# The Demand Curve

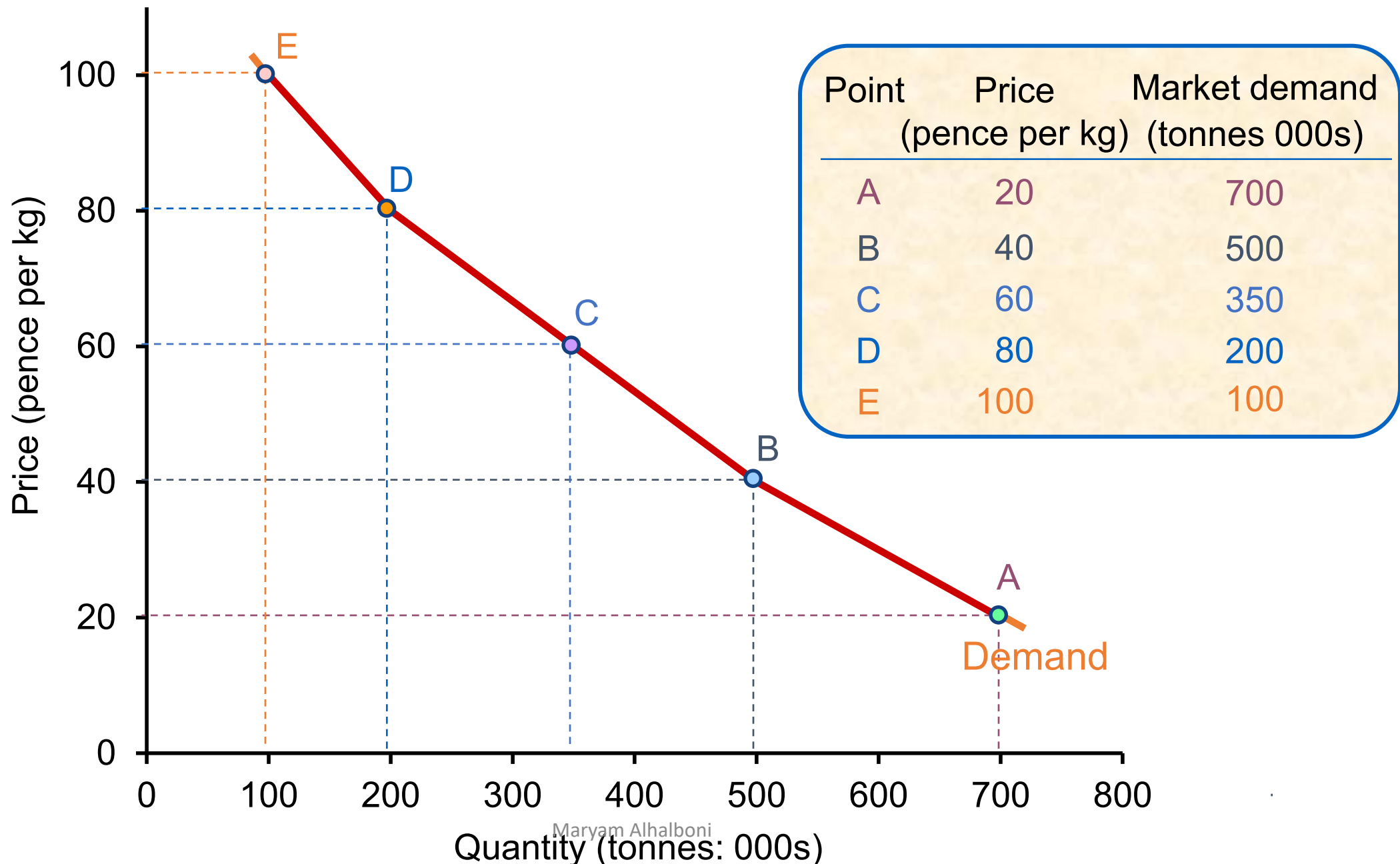
- The demand curve translates the quantity of a product (e.g., potatoes) purchased at different price levels over a specific period into a visual format.
  - Vertical Axis (Y): Represents Price (P).
  - Horizontal Axis (X): Represents Quantity Demanded (Q).
- Downward Slope: Illustrates the inverse relationship between price and quantity.

# The demand curve:

## the demand for potatoes (monthly)

|   | (1)<br>Price<br>(pence per kg) | (2)<br>Tracey's<br>demand<br>(kg) | (3)<br>Darren's<br>demand<br>(kg) | (4)<br>Total market<br>demand<br>(tonnes: 000s) |
|---|--------------------------------|-----------------------------------|-----------------------------------|-------------------------------------------------|
| A | 20                             | 28                                | 16                                | 700                                             |
| B | 40                             | 15                                | 11                                | 500                                             |
| C | 60                             | 5                                 | 9                                 | 350                                             |
| D | 80                             | 1                                 | 7                                 | 200                                             |
| E | 100                            | 0                                 | 6                                 | 100                                             |

# Market demand for potatoes (monthly)



# Pricing Strategies and Promotions

- There is a negative relationship between price and quantity.
- Businesses often prefer "clever promotions" over direct (overt) price cuts to avoid negative consequences.
- Risks of direct price reductions:
  - **Price Wars:** Competitors may retaliate with even lower prices, destroying profit margins for the whole industry.
  - **Quality Signaling:** Consumers might perceive a lower price as a sign of inferior quality or a "cheap" brand image.

# Pricing Strategies and Promotions

## The Power of 'BOGOF' (Buy One Get One Free):

- Allows the "published" or "headline" price to remain stable while effectively cutting the cost for the consumer by 50%.
- Used to aggressively boost sales volume and capture market share without permanently devaluing the product.

## End-of-Season Sales:

- Sales are a practical application of the demand curve in action.
- Retailers lower prices to a level where consumers are finally willing and able to "shift" remaining stock that didn't sell at the original higher price.

# Determinants of Demand

Beyond price, several factors influence how much consumers will buy:

- **Tastes and Preferences:**

- Influenced by
  - Fashion & Trends: What is "in" today may be "out" by next season.
  - Technological Progress: Newer, better features (e.g., high-definition cameras on smartphones) render older technology less desirable.
  - Advertising: A powerful tool used by companies to actively shape what consumers find appealing.
- An improvement in tastes (e.g., a new fashion trend) shifts the demand curve to the right, meaning more is sold at every price.
- A reduction in backing for a product shifts the curve to the left.

# Determinants of Demand

- **Substitute Goods:** Goods used in place of another (e.g., tea vs. coffee).
  - If the price of tea rises, demand for coffee increases.
  - If the price of a substitute falls, the demand curve for the original product shifts left.
  - Conversely, if the substitute's price rises, demand shifts right.
- **Complementary Goods:** Goods used together (e.g., consoles and games).
  - If the price of an Xbox rises, the demand for Xbox games falls.

# Determinants of Demand

- **Income Levels:**
  - **Normal Goods:**
    - Products for which demand increases as consumer income rises (e.g., Branded goods).
    - A rise in income shifts the demand curve to the right.
    - During a recession (falling income), the curve shifts left.
  - **Inferior Goods:**
    - Demand falls as income increases (e.g., "value" brands or bus travel) as people switch to higher-quality options.
- **Income Distribution:** Shifting wealth from poor to rich increases demand for luxury items.

# Example: Supermarket Trends & Choice Reduction

- Grocery stores provide a clear view of income effects by placing premium brands (Normal) and value lines (Inferior) side-by-side.
- Post-Pandemic Shift: High inflation and the cost-of-living crisis have forced a change in supermarket strategy.
- Reduced Product Ranges: Supermarkets are streamlining to focus on the most popular, high-volume lines.
  - Since 2019, UK supermarkets reduced overall product ranges by 9%.
  - Asda reduced its range by 26%, reflecting a move toward more concentrated demand as consumers' purchasing power fluctuates.
- Income changes do not just change how much we buy, but also the variety of choices available on the shelf.

# Determinants of Demand

- **Expectations:** Consumers' current buying behavior is heavily influenced by what they think will happen to prices later.
  - Expectation of Falling Prices:
    - If you believe a smartphone will be cheaper in six months, you delay your purchase.
    - Current demand shifts left while future demand shifts right.
  - Expectation of Rising Prices:
    - Common in markets like property, shares, or oil. If you think prices will climb, you buy now to save money later.
    - Current demand shifts right as consumption is brought forward, while future demand shifts left.

# Determinants of Demand

- **Advertising**

- 1. **The Informational Role:**

- Informs consumers about new products, specific features, or lower prices.
    - **Effect:** Increases demand simply by making the product's existence known.

- 2. **The Persuasive Role (Shaping Tastes):**

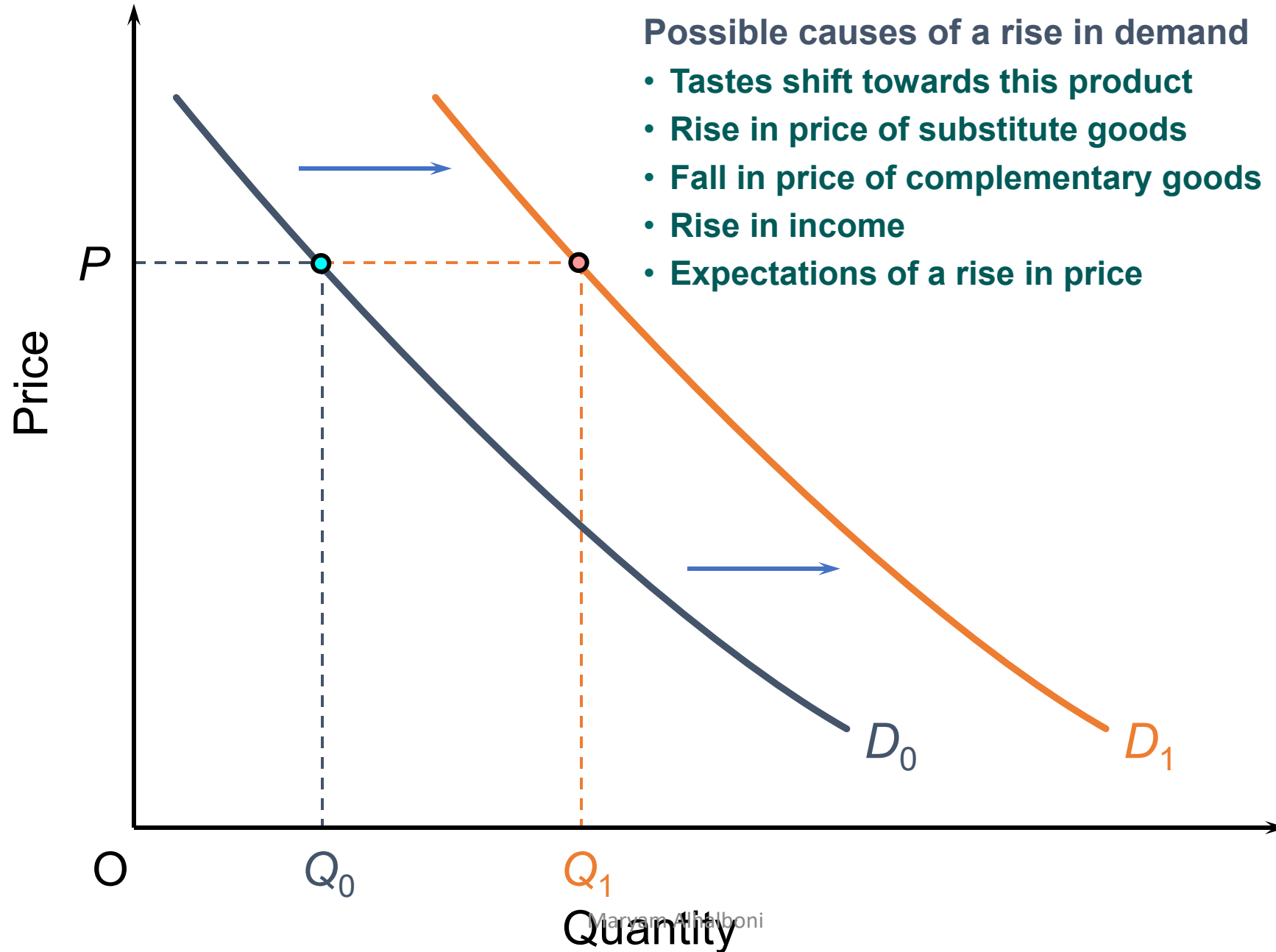
- Focuses on lifestyle and differentiation rather than just facts.
    - **Celebrity Endorsements:** Using famous figures to suggest that a product defines a desirable identity or status.
    - **Effect:** Aims to create a perceived difference between identical functional products (e.g., two phones with the same specs, but only one is "cool").

- **The Goal:** Every advertisement seeks to shift the firm's demand curve **out (right)** while simultaneously shifting the competitors' curves **in (left)**.

# Movements vs. Shifts in Demand

- Movement Along the Curve: Occurs only when the price of the good changes. This is called a change in the quantity demanded.
- Shift of the Curve: Occurs when a factor other than price changes (e.g., a rise in income).
  - Shift to the Right: Increase in demand (people buy more at every price).
  - Shift to the Left: Decrease in demand (people buy less at every price).
  - This is referred to as a change in demand.

# An increase in demand



**Q** Which way will the market demand for petrol shift if the price of cars rises?

A. Right

☒ B. Left

C. No shift (movement along the curve)

**Q** Which way will the market demand for petrol shift if petrol becomes more expensive?

A. Right

B. Left

☒ C. No shift (movement along the curve)

# Do Higher Prices Attract Higher Demand?

## The Veblen Effect and the Luxury Paradox

- The Common Misconception: It is often assumed that for luxury items (like designer perfumes or bags), higher prices create demand.
- The Economic Reality: Even for the ultra-rich, the Law of Demand holds true. Because of budget constraints, a higher price for one luxury item (e.g., designer clothes) means less money is available for others (e.g., private jets and champagne).
- Affordability vs. Exclusivity: Even if your budget is £500,000, an increase in the price of clothes forces a choice: buy fewer clothes, or cut back on other luxuries.

# Explaining the "Positive" Relationship

- Signaling and Image: For brands like Louis Vuitton, a high price acts as a signal of high quality and exclusivity. A low price would damage the brand's "exclusive" image.
- Tastes and Preferences (The Shift): The reason we sometimes see more being bought at higher prices is not a movement along a curve, but a shift of the curve.
  - A high price attracts a specific segment of consumers with a "taste for exclusivity," shifting the demand curve to the right.
  - A discount might attract mass-market consumers but cause the "exclusive" consumers to lose interest, shifting the original demand curve to the left.

# The Working of Competitive Markets- Supply

- As the price of a good rises, the quantity supplied by firms also rises (a positive relationship).
- Reasons for the Upward Slope:
  - Rising Costs: Producing more often leads to higher marginal costs (e.g., paying overtime or using less fertile land). Higher prices are needed to cover these extra expenses.
  - Profitability: Higher prices make a product more attractive to produce, encouraging firms to switch resources away from less profitable goods.
- Short Run vs. Long Run:
  - Short Run: Existing firms increase production using current capacity.
  - Long Run: If prices remain high, new firms enter the market, further increasing total market supply.

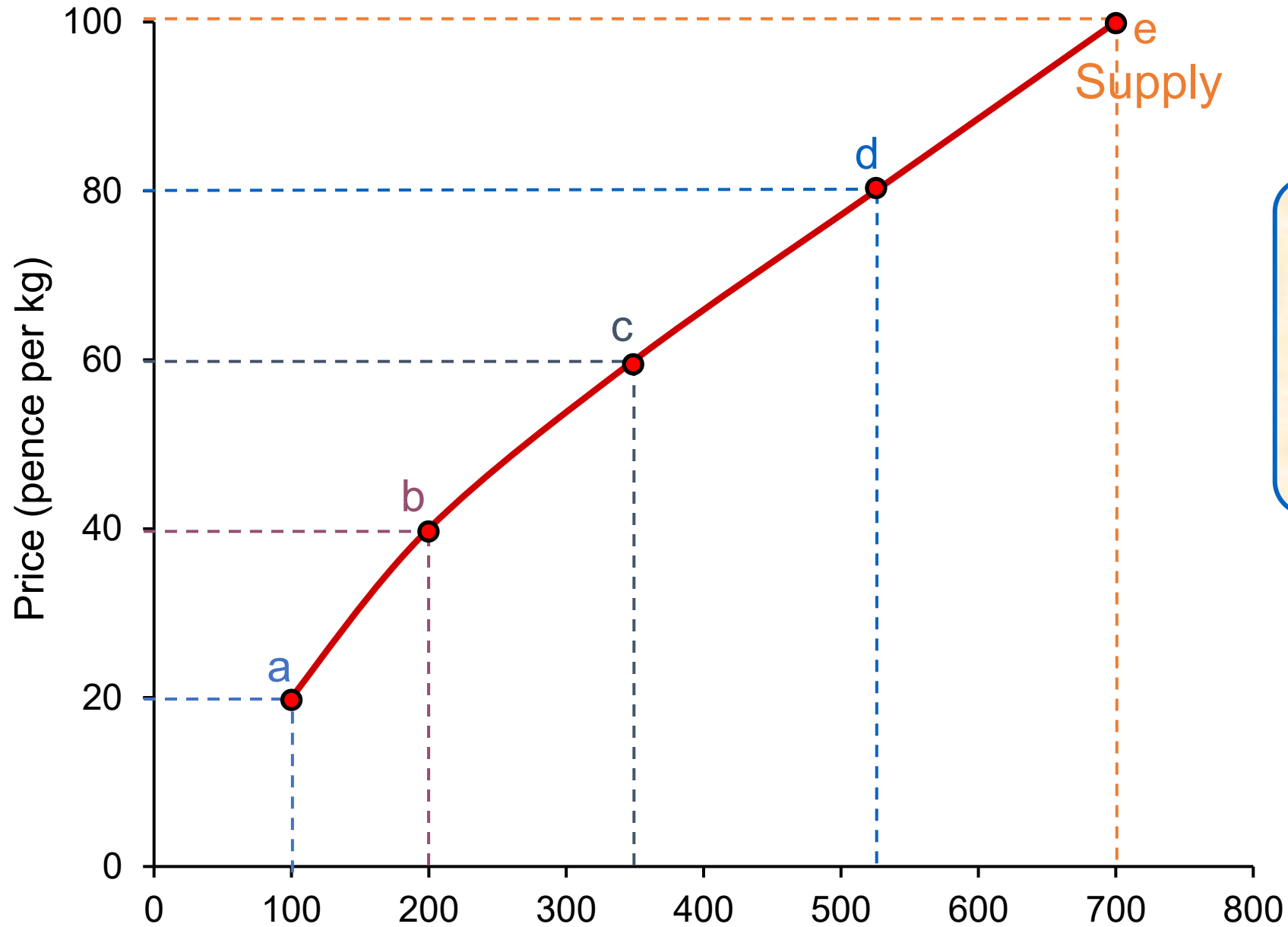
# The Supply Curve

- The Supply curve translates the quantities of a product (e.g., potatoes) that producers are willing and able to sell at different prices into a visual format.
  - Vertical Axis (Y): Represents Price (P).
  - Horizontal Axis (X): Represents Quantity Supplied (Q).
- A change in the price of the good itself causes a movement along the curve.
- While typically upward sloping, supply curves can be vertical or horizontal depending on how quickly firms can respond to price changes over a specific time period.

# The supply curve: the supply of potatoes (monthly)

|   | Price of<br>potatoes<br>(pence per kg) | Farmer X's<br>supply<br>(tonnes) | Total market<br>supply<br>(tonnes: 000s) |
|---|----------------------------------------|----------------------------------|------------------------------------------|
| a | 20                                     | 50                               | 100                                      |
| b | 40                                     | 70                               | 200                                      |
| c | 60                                     | 100                              | 350                                      |
| d | 80                                     | 120                              | 530                                      |
| e | 100                                    | 130                              | 700                                      |

# Market supply of potatoes (monthly)



|   | <i>P</i> | <i>Q</i> |
|---|----------|----------|
| a | 20       | 100      |
| b | 40       | 200      |
| c | 60       | 350      |
| d | 80       | 530      |
| e | 100      | 700      |

# Determinants of Supply

Several factors beyond price influence the total amount firms bring to market:

- **Costs of Production:** Higher costs (wages, raw materials) reduce profit margins, causing firms to cut back supply or switch products.
- **Profitability of Alternative Products (Substitutes in Supply):** If a different product becomes more profitable, firms will switch resources, reducing the supply of the original good.
- **Joint Supply:** When producing one good automatically results in another (e.g., refining crude oil into petrol also produces diesel and paraffin).

# Determinants of Supply

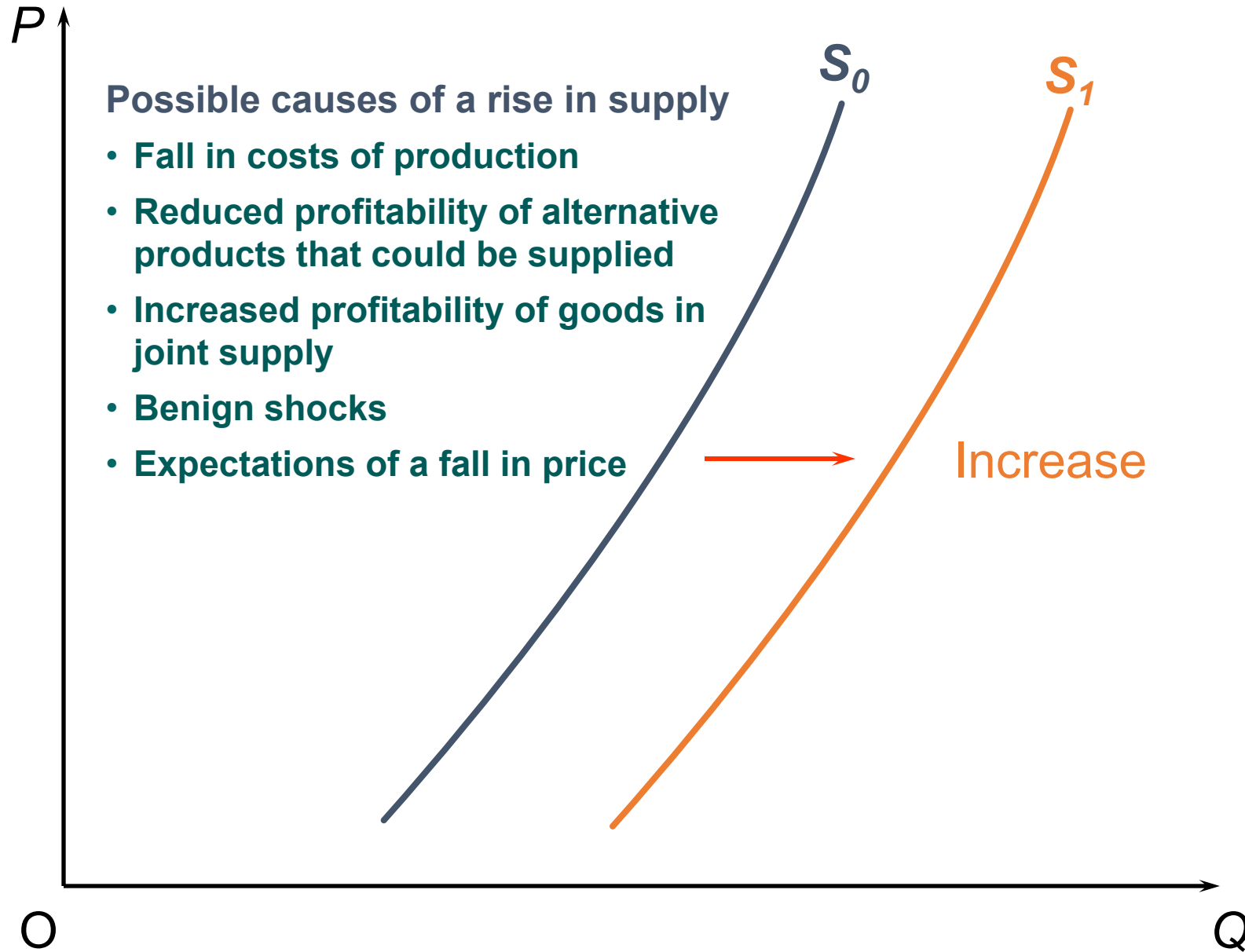
Several factors beyond price influence the total amount firms bring to market:

- **Random Shocks:** Unpredictable events such as extreme weather, crop diseases, wars, or industrial disputes can significantly disrupt supply.
- **Producers' Aims:** A firm focused on market share (sales maximization) will supply a different quantity than one focused purely on profit maximization.
- **Expectations:** If prices are expected to rise later, firms may temporarily "stockpile" goods, reducing current supply.
- **Number of suppliers:** An increase in the number of suppliers entering the market will shift the total supply upward.

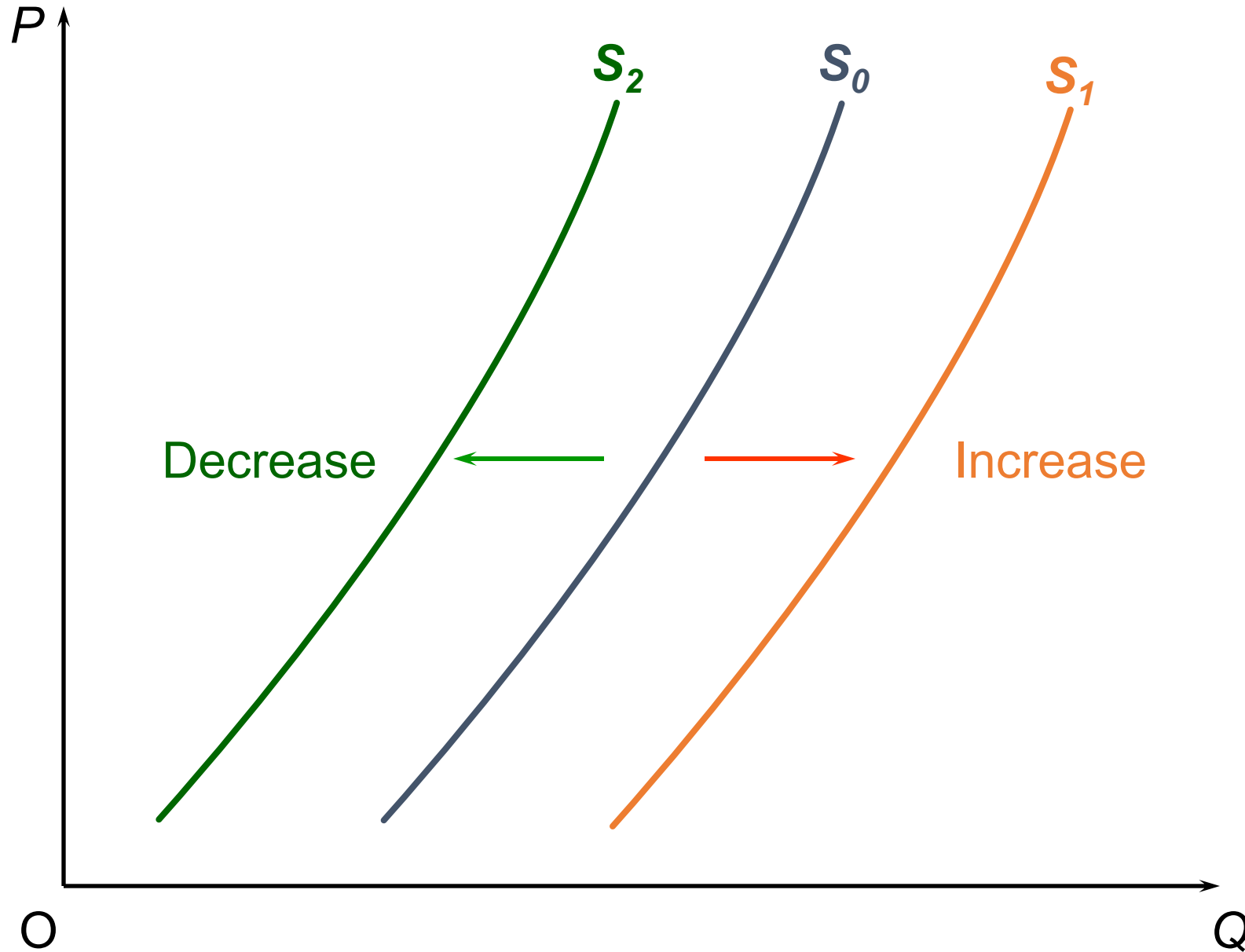
# Movements vs. Shifts in Supply

- Change in "Quantity Supplied": A movement along the existing curve caused only by a change in the price of the product.
- Change in "Supply": A shift of the entire curve caused by a change in any other determinant (e.g., a decrease in production costs).
- Direction of Shifts:
  - Rightward Shift ( $S_0$  to  $S_1$ ): Represents an increase in supply (more provided at every price).
  - Leftward Shift ( $S_0$  to  $S_2$ ): Represents a decrease in supply (less provided at every price).

# Shifts in the supply curve



# Shifts in the supply curve



**Q** Which way will the market supply of pizzas shift if the price of flour falls?

**A.** Right

**B.** Left

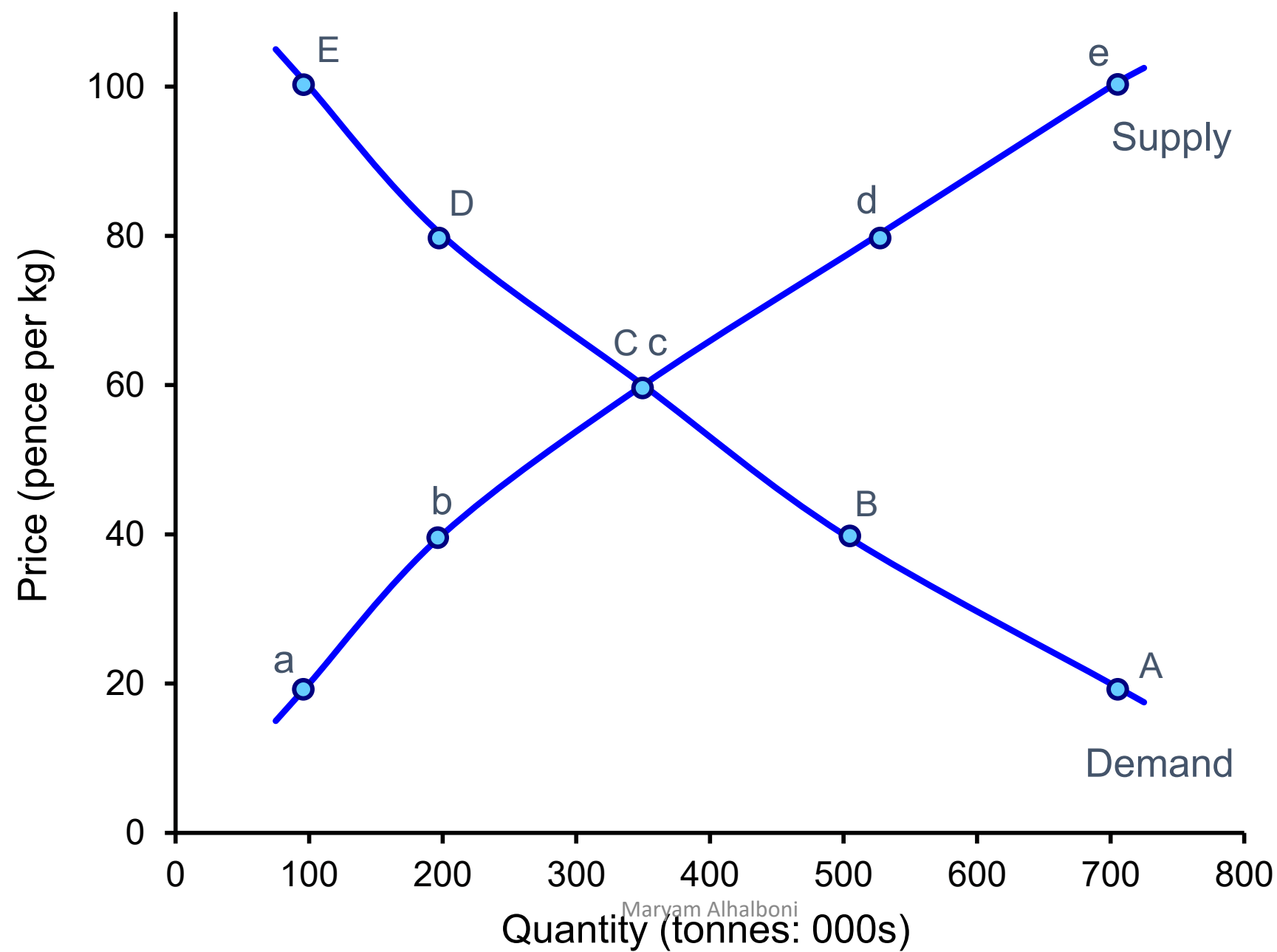
**C.** No shift (movement along the curve)

# The Working of Competitive Markets-Business in a Competitive Market

# Market Equilibrium – Bringing Demand and Supply Together

- Equilibrium: The point where the plans of consumers (Demand) and the plans of producers (Supply) meet.
- Dealing with Imbalance:
  - Shortage (Price < Equilibrium): At 20p, demand exceeds supply. Consumers bid up the price, and producers increase output until the shortage is gone.
  - Surplus (Price > Equilibrium): At 100p, supply exceeds demand. Farmers must lower prices to clear excess stock, leading to a price drop.
- The Market Clearing Price ( $P_e$ ): The sustainable price where demand equals supply (e.g., 60p). At this point, there is no shortage and no surplus.
- On a graph, equilibrium is found exactly where the demand and supply curves intersect.

# The determination of market equilibrium (potatoes: monthly)

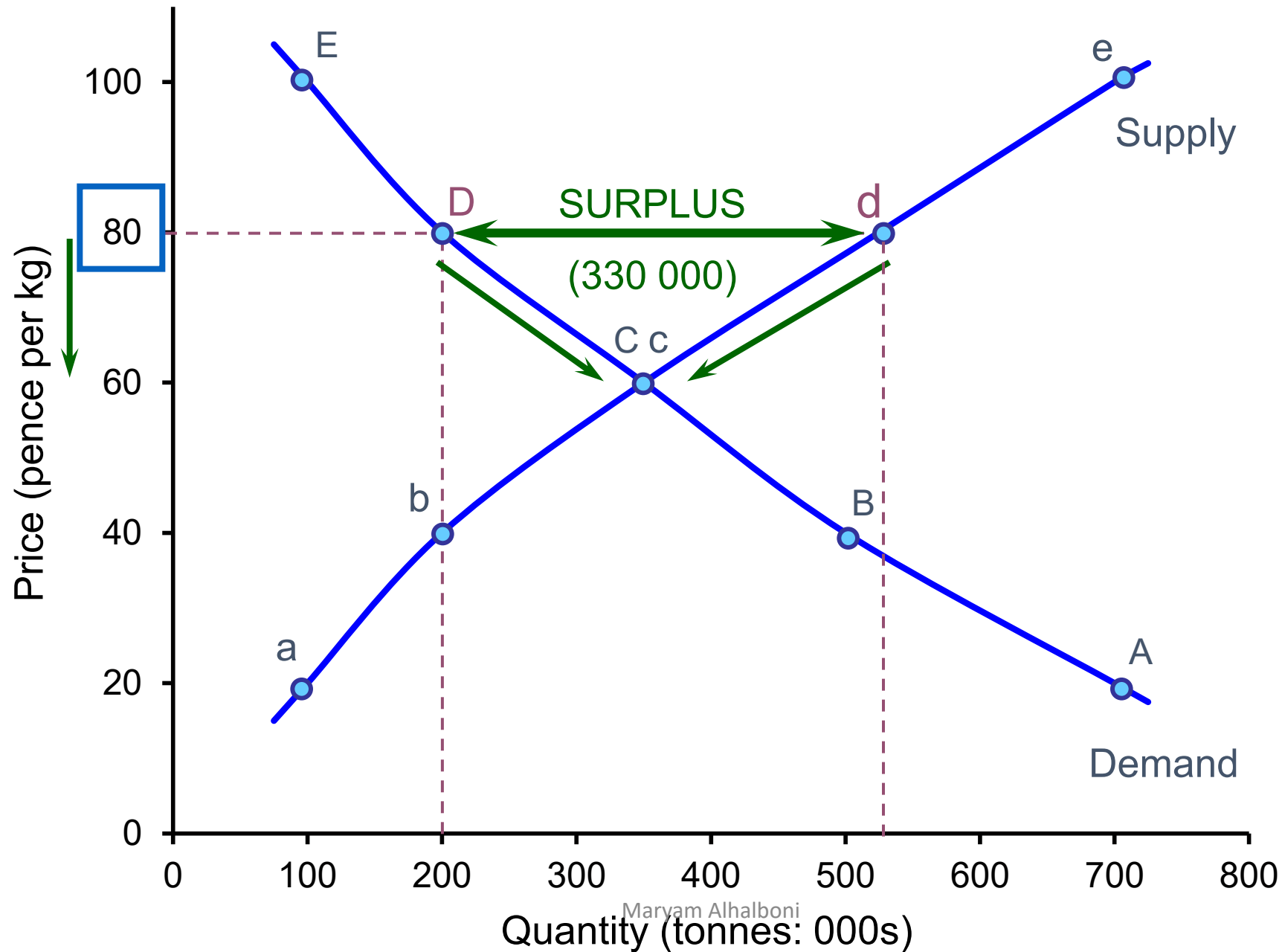


# Market Disequilibrium – Surpluses and Shortages

Handling a Surplus (Price > Equilibrium):

- At a price of 80p, a surplus of 330,000 tonnes occurs (where  $S > D$ ).
- Because more is supplied than consumers are willing to buy, the price is forced downward.
- The Adjustment: As the price falls toward 60p, we see a movement along the demand curve (D to C) and the supply curve (d to c) until the market clears.

# The determination of market equilibrium (potatoes: monthly)



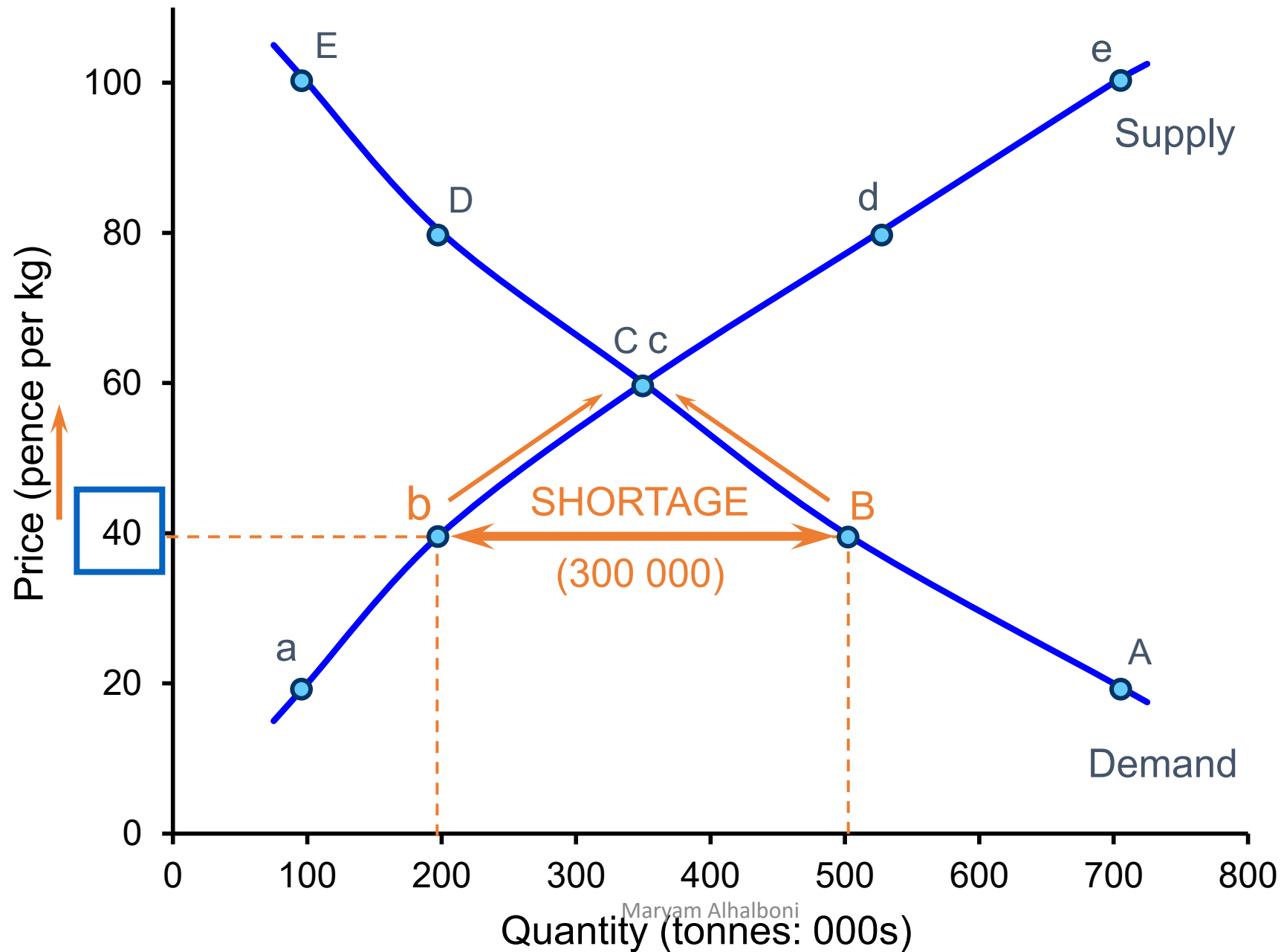
# Market Disequilibrium – Surpluses and Shortages

Handling a Shortage (Price < Equilibrium):

- At a price of 40p, a shortage of 300,000 tonnes occurs (where  $D > S$ ).
- Competition among buyers drives the price upward.
- The Adjustment: As the price rises toward 60p, we see a movement along the supply curve (b to c) and the demand curve (B to C).

The Point of Balance: Point Cc represents the stable equilibrium where the quantity demanded perfectly matches the quantity supplied.

# The determination of market equilibrium (potatoes: monthly)

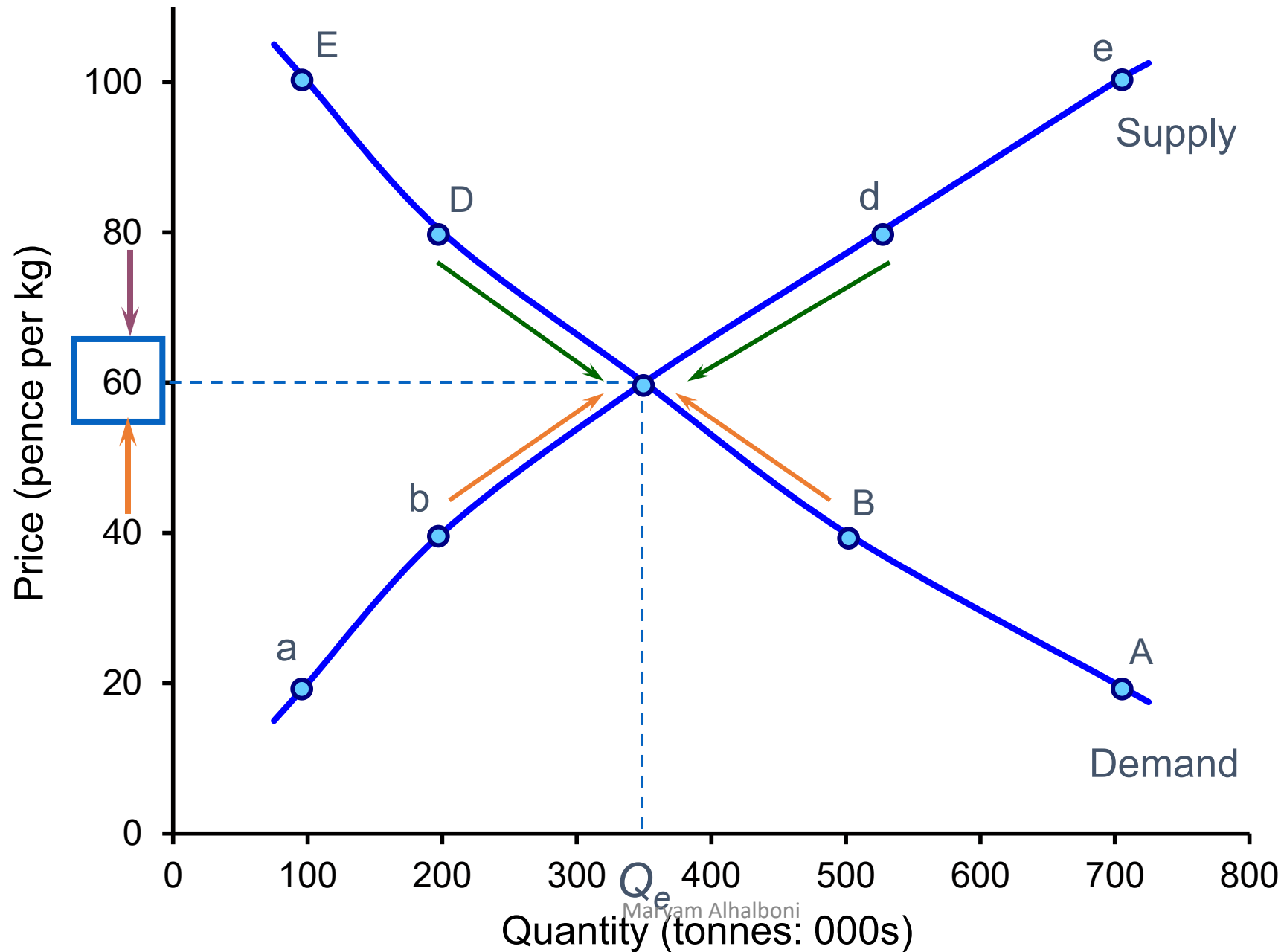


# Equilibrium price and output:

the market demand and supply of potatoes (monthly)

| Price of potatoes<br>(pence per kilo) | Total market demand<br>(tonnes: 000s) | Total market supply<br>(tonnes: 000s) |
|---------------------------------------|---------------------------------------|---------------------------------------|
| 20                                    | 700 (A)                               | 100 (a)                               |
| 40                                    | 500 (B)                               | 200 (b)                               |
| 60                                    | 350 (C)                               | 350 (c)                               |
| 80                                    | 200 (D)                               | 530 (d)                               |
| 100                                   | 100 (E)                               | 700 (e)                               |

# The determination of market equilibrium (potatoes: monthly)



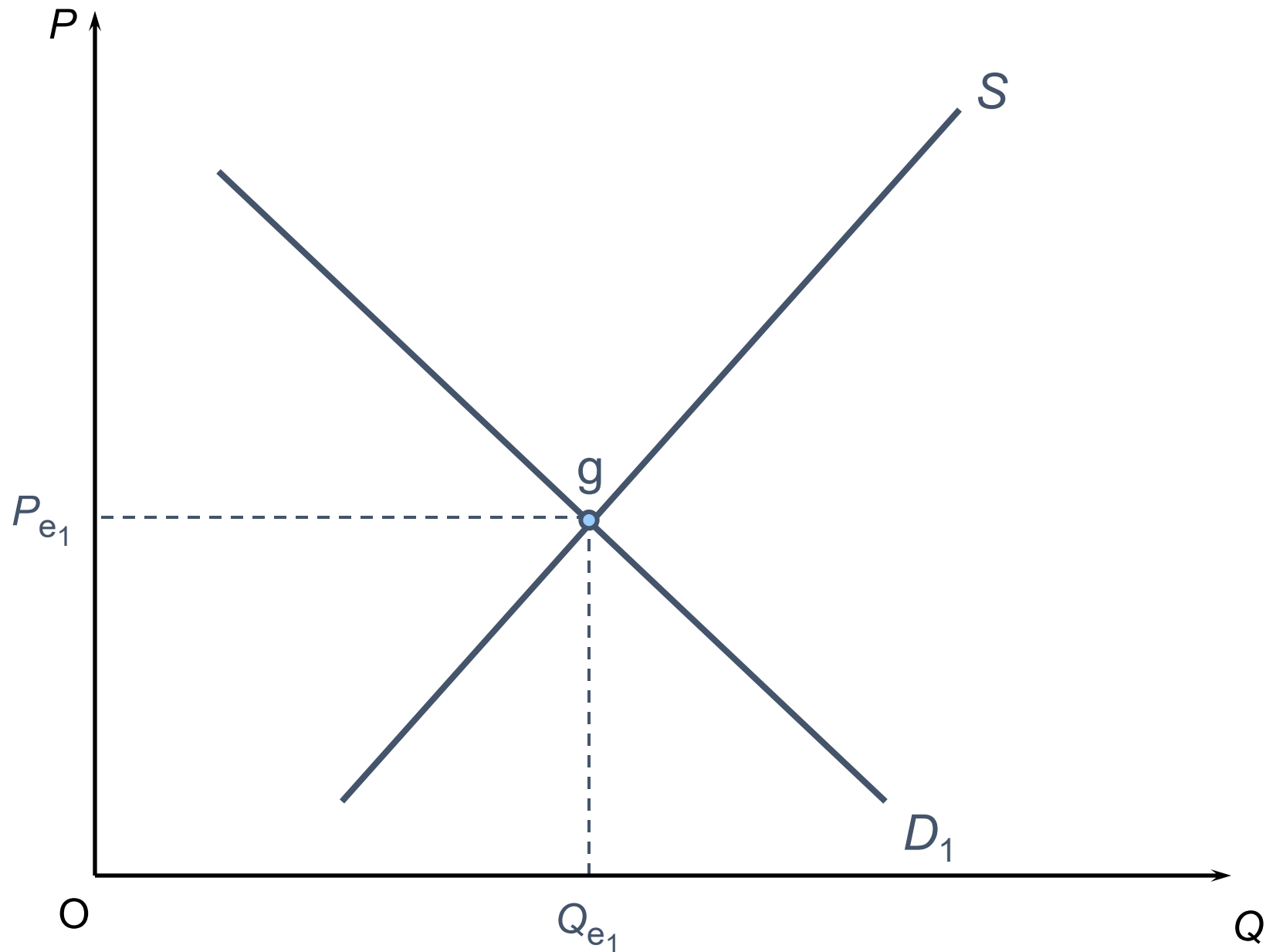
# Movement to a New Equilibrium – Change in Demand

- Equilibrium only remains stable if the curves stay put. If a determinant other than price changes, a curve shifts and a new equilibrium forms.

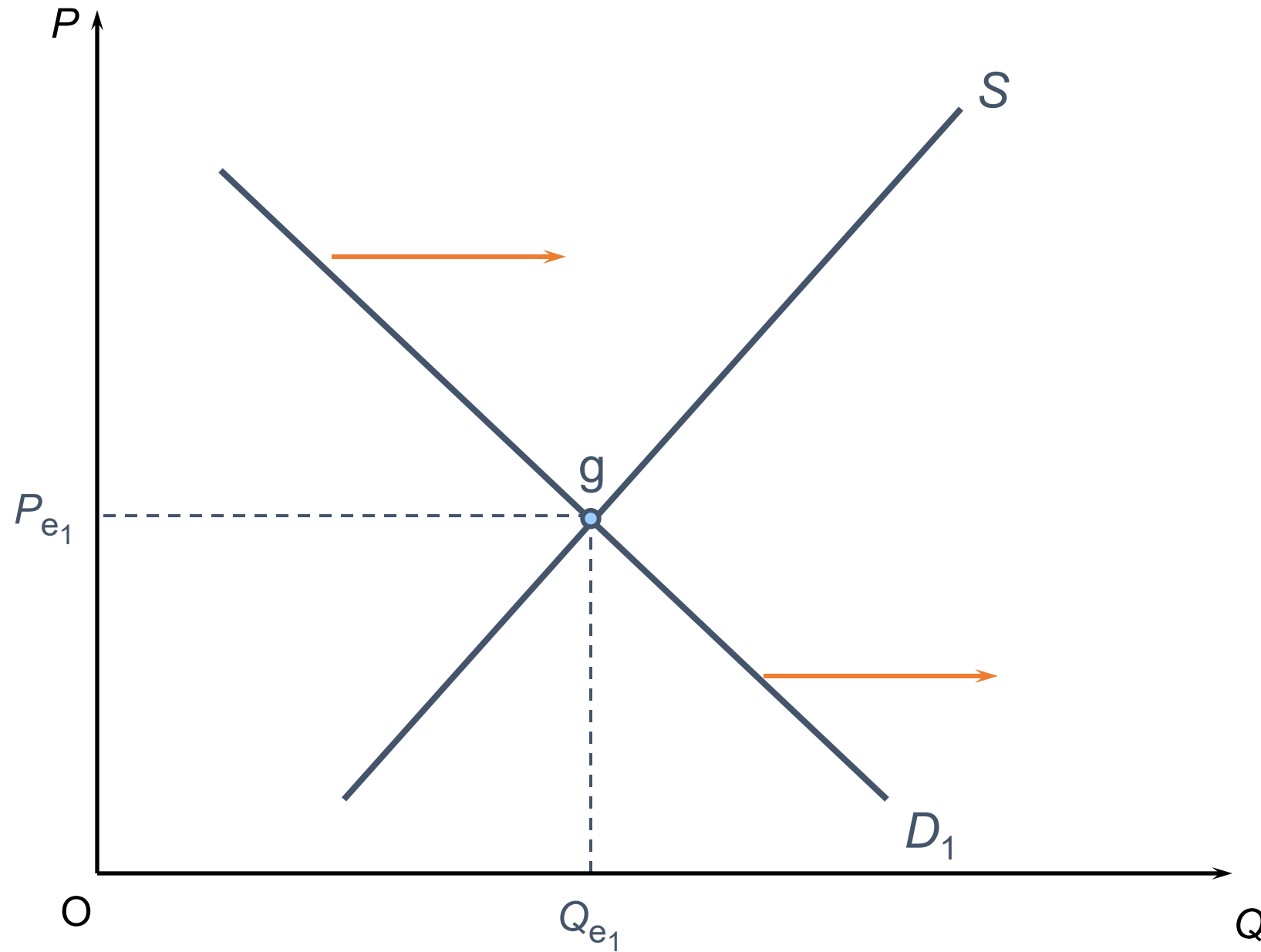
## The Impact of a Demand Shift:

- Suppose consumer incomes rise, shifting the demand curve to the right ( $D_1$  to  $D_2$  ).
- At the original price, a shortage ( $Q_{e_1} - Q_{e_2}$ ) is created.
- Price rises to  $P_{e_2}$  and the equilibrium quantity rises to  $Q_{e_2}$ .
- A shift in the Demand curve results in a movement along the Supply curve to reach the new intersection.

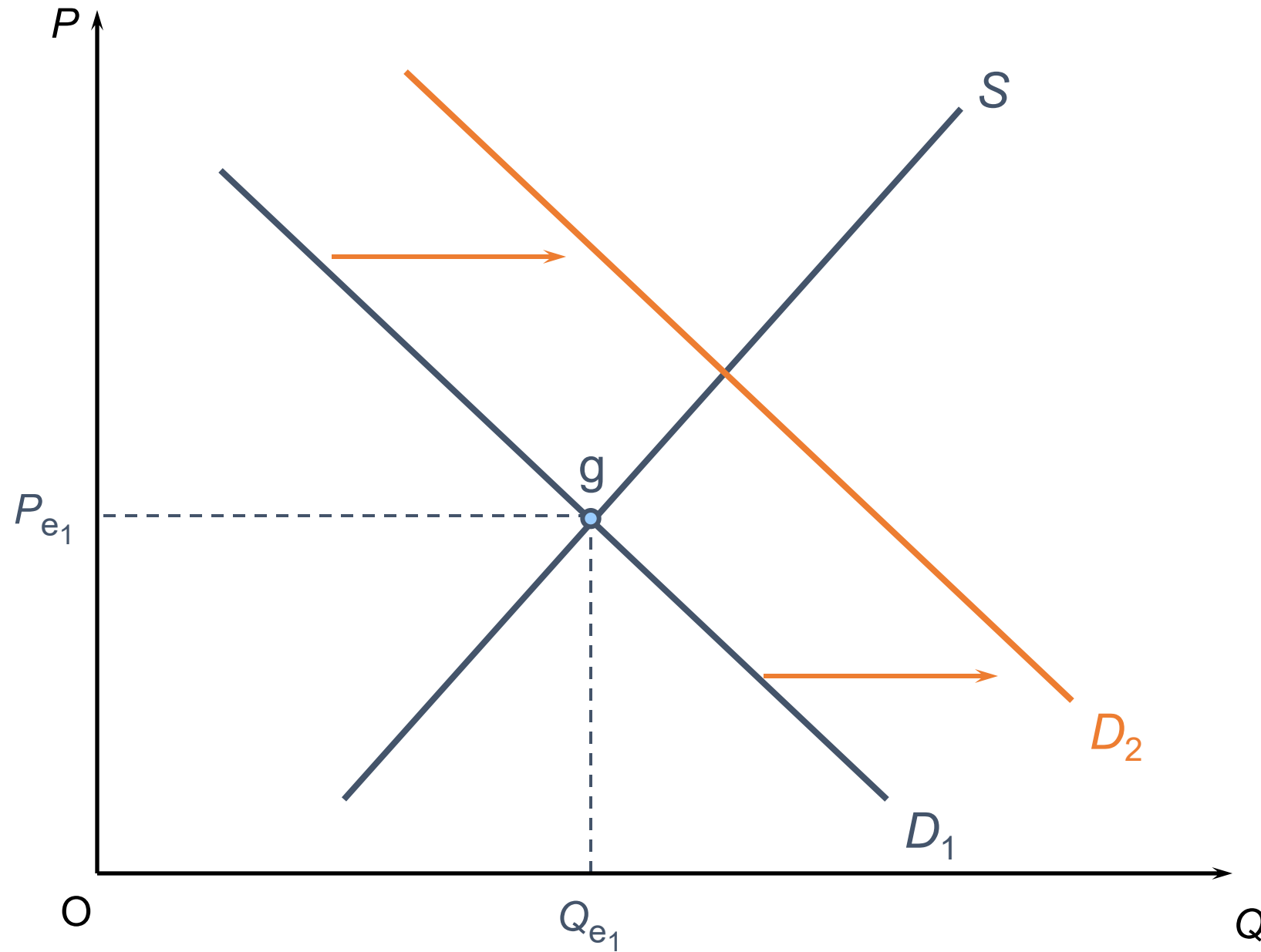
Effect of a shift in the demand curve



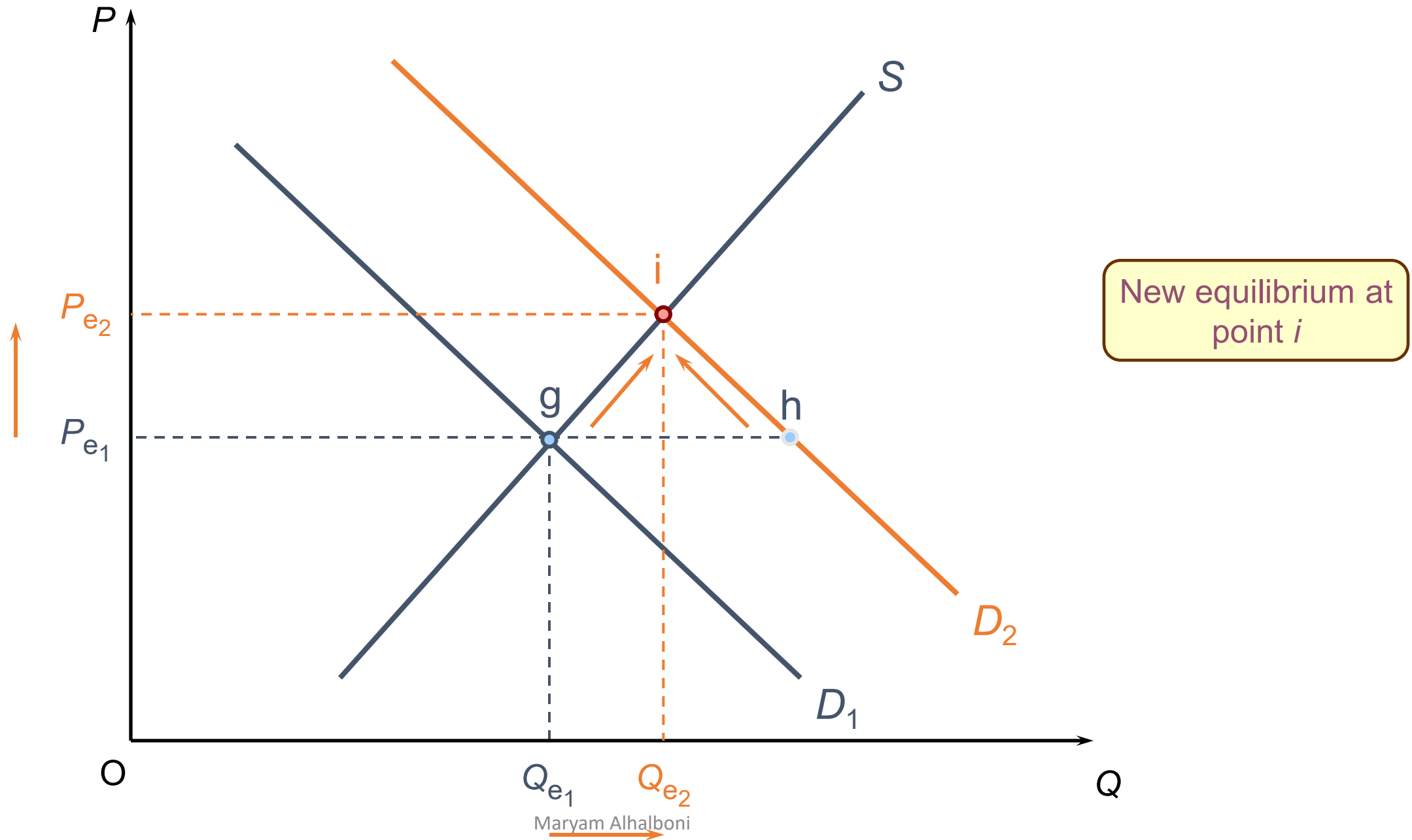
## Effect of a shift in the demand curve



## Effect of a shift in the demand curve



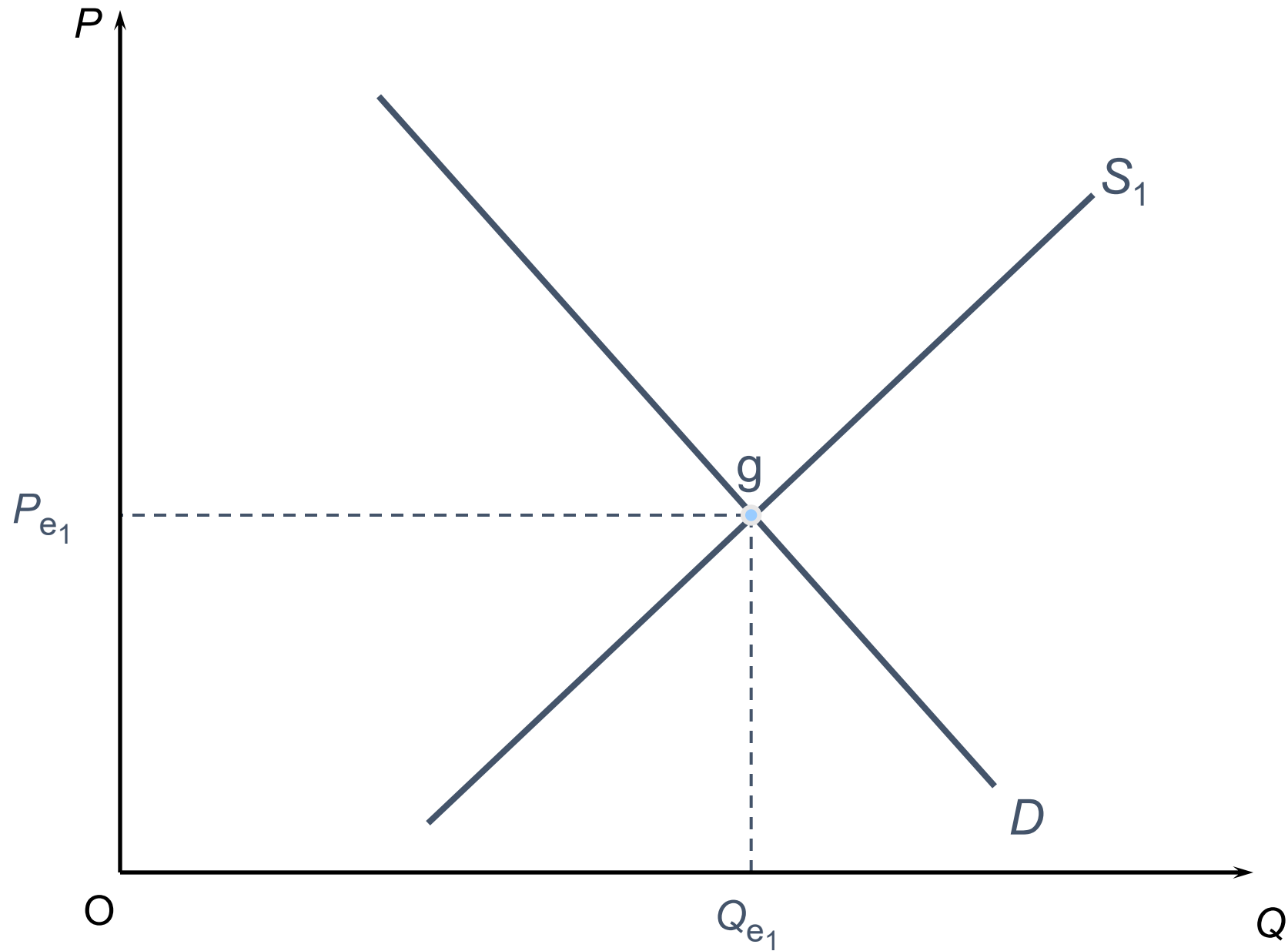
## Effect of a shift in the demand curve



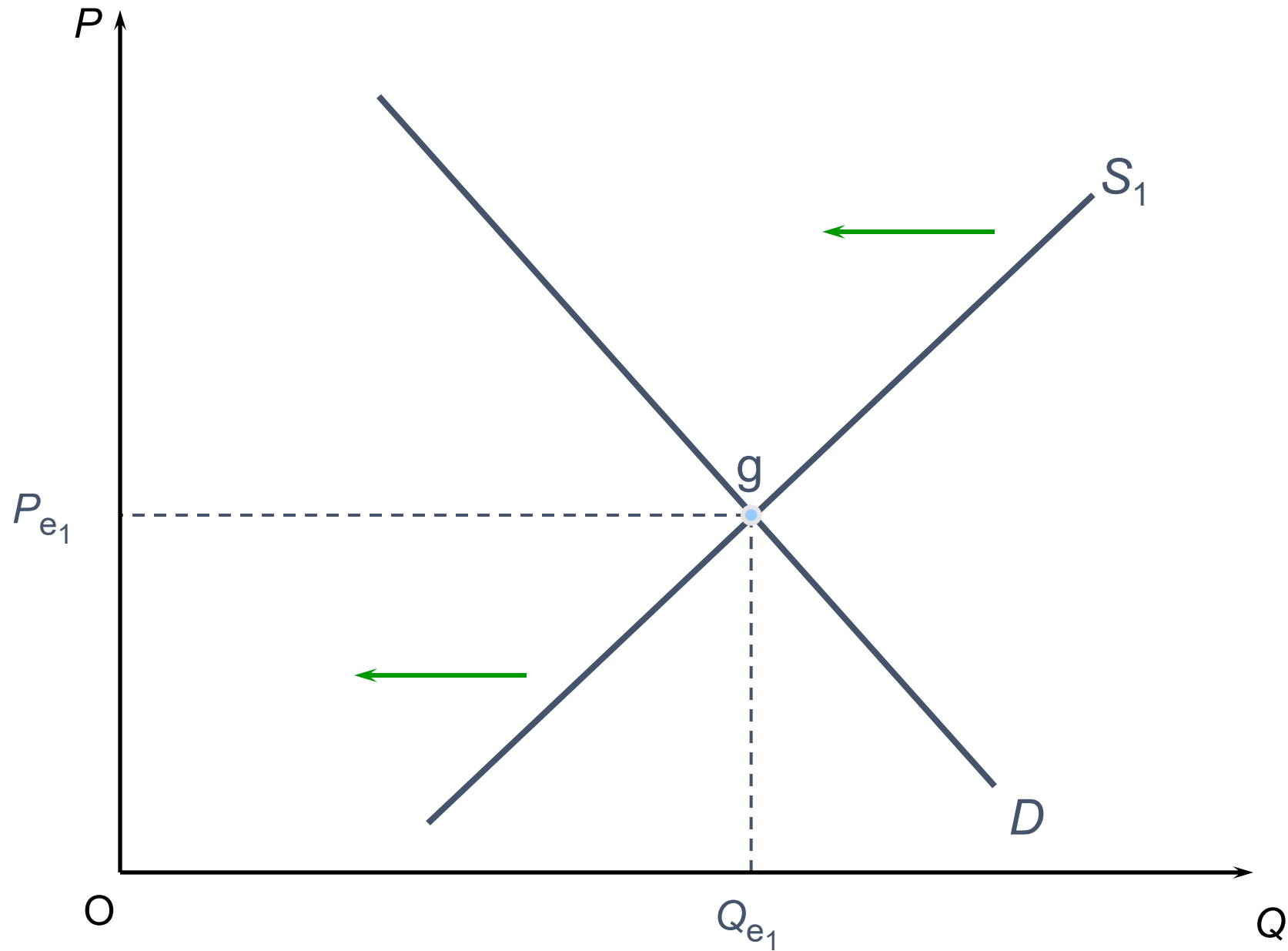
# Movement to a New Equilibrium – Change in Supply

- Suppose production costs rise, shifting the supply curve to the left ( $S_1$  to  $S_2$ ).
- At the old price  $P_{e_1}$ , a shortage ( $Q_{e_1} - Q_{e_2}$ ) occurs because producers cannot afford to supply as much.
- Price rises to  $P_{e_3}$  to cover the higher costs, and the equilibrium quantity falls to  $Q_{e_3}$  as consumers buy less at the higher price.
- A shift in the Supply curve results in a movement along the Demand curve to reach the new intersection
- If both demand and supply determinants change at the same time, both curves will shift. The new equilibrium is simply the intersection of the two new curves.

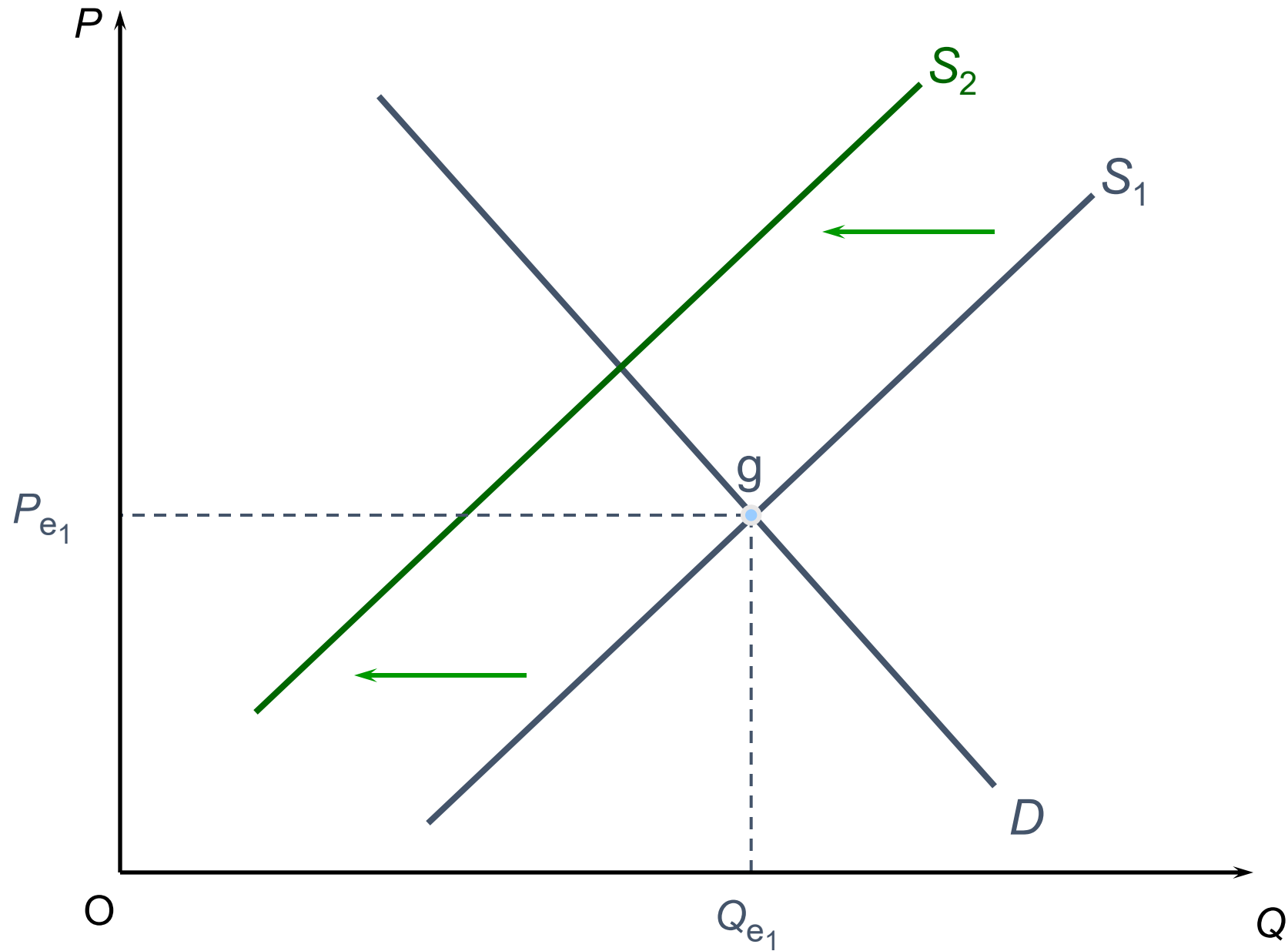
# Effect of a shift in the supply curve



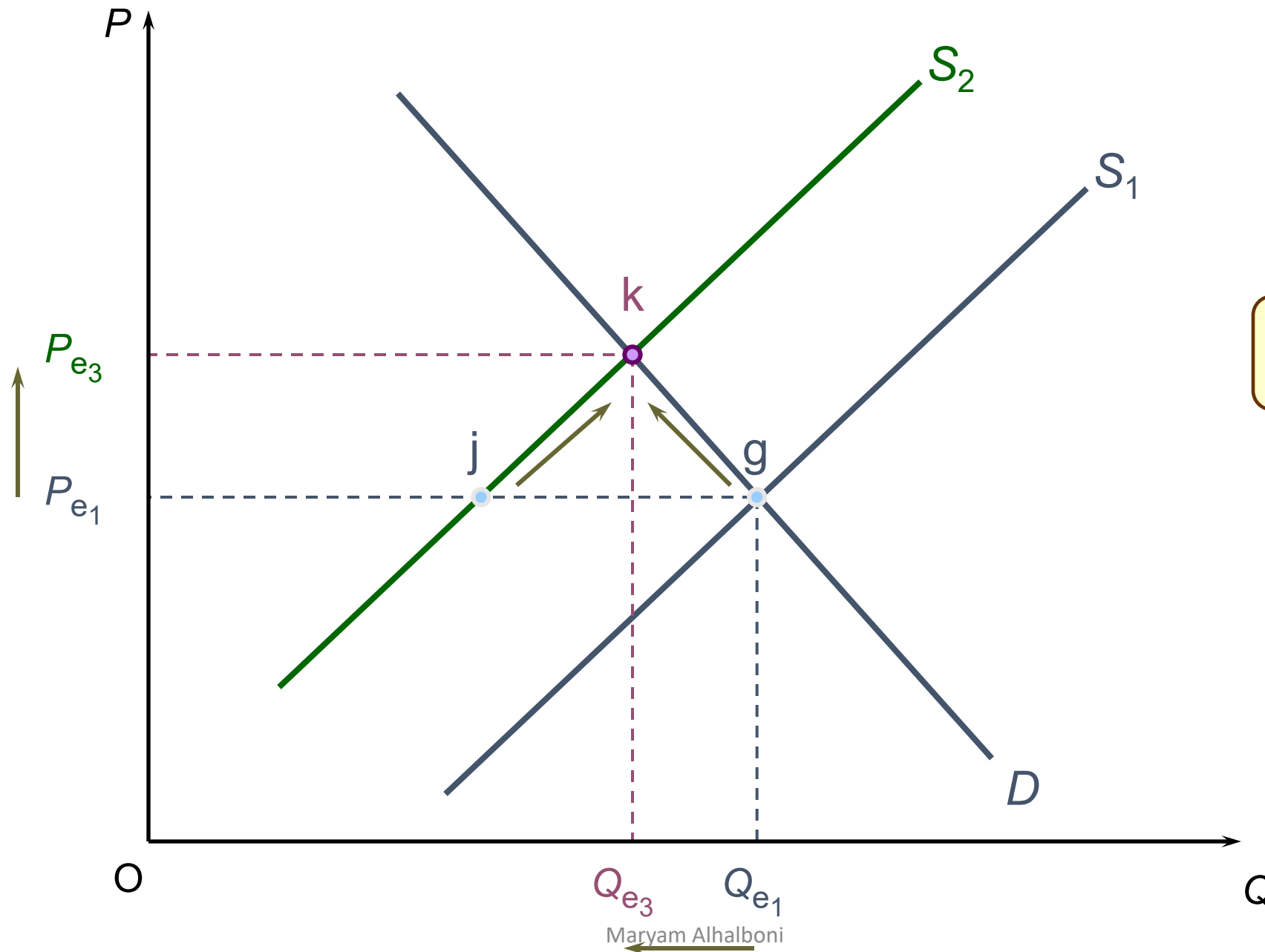
# Effect of a shift in the supply curve



# Effect of a shift in the supply curve

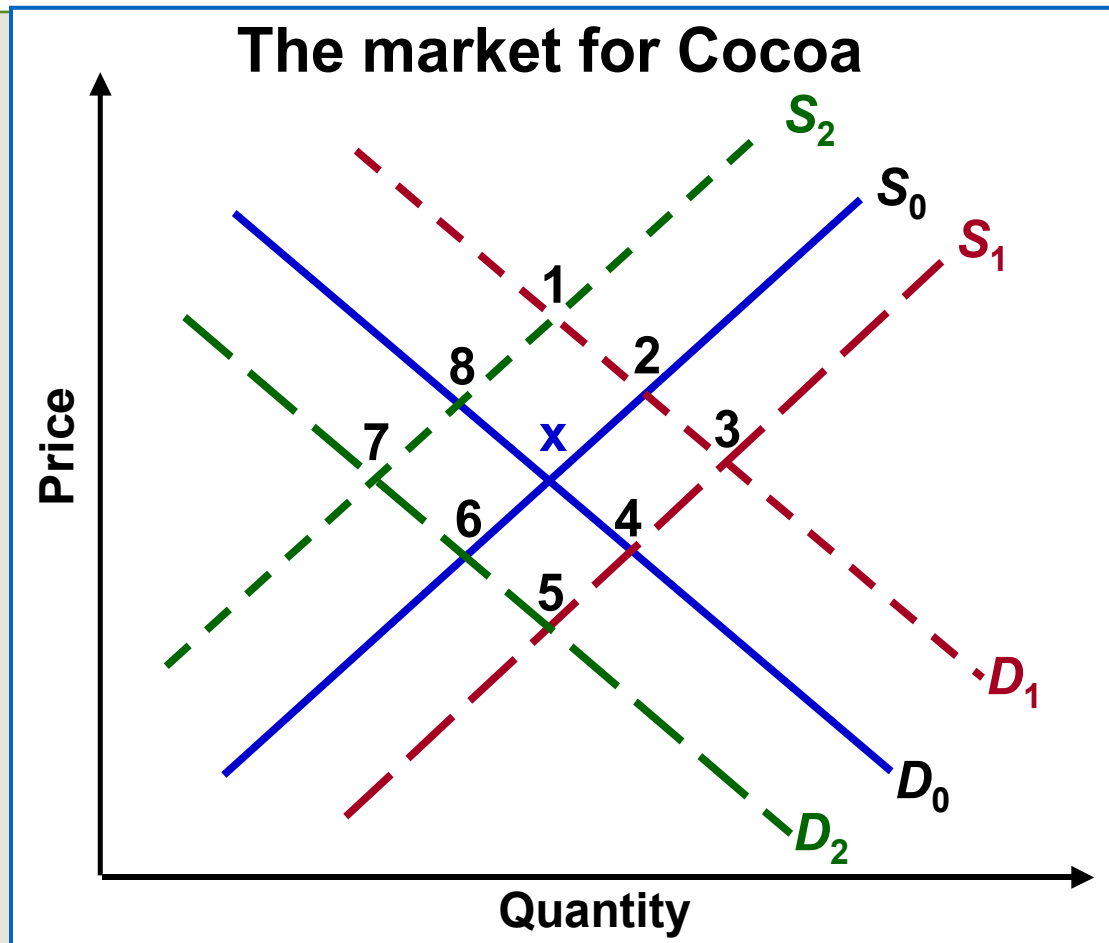


# Effect of a shift in the supply curve

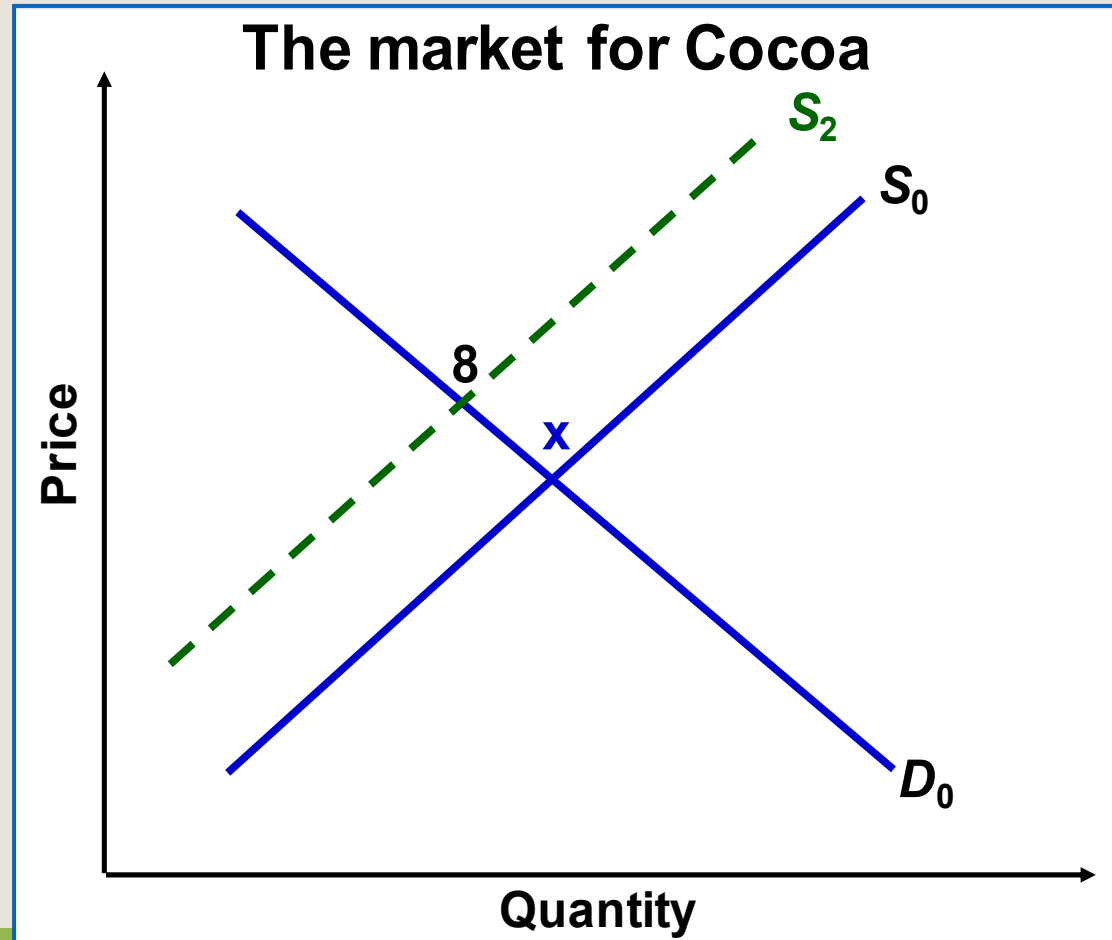


New equilibrium at point  $k$

**Q** The diagram shows the market for cocoa. Equilibrium is currently at point x. To which equilibrium point (1, 2, 3, 4, 5, 6, 7 or 8) will the market move if there is a rise in the cost of producing cocoa?



**Q** The diagram shows the market for cocoa. Equilibrium is currently at point x. To which equilibrium point (1, 2, 3, 4, 5, 6, 7 or 8) will the market move if there is a rise in the cost of producing cocoa?



# Questions